Arash Jalil Khabbazi, MS

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Profile

I develop advanced control strategies such as reinforcement learning and model predictive control, and apply machine learning to make buildings, distributed energy resources, and other AI-driven technologies smarter and more efficient.

Education

Purdue University, *IN, USA* 2023 – 2027
PhD in Mechanical Engineering — **GPA:** 3.86/4.0
Minor in Computational Science and Engineering

University of British Columbia, *BC, Canada* 2021 – 2023
MS in Mechanical Engineering — **GPA:** 4.0/4.0 (94%)

University of Tabriz, *EA, Iran* 2016 – 2020
BS in Mechanical Engineering — **GPA:** 4.0/4.0 (19.12/20) — **Rank:** 1/155+

Experience

Applied ML Intern, Tesla, *Reno, NV, USA* 05/2025 – 08/2025

- Built supervised ML models (XGBoost, neural nets, SVR, and ensembles) for defect detection and quality assessment.
- Contributed to deployment of a thermal imaging system for non-destructive inspection in drive-unit manufacturing.
- Developed data pipelines in MySQL and engineered features to extract high-signal inputs for machine learning models.
- Collaborated on projects projected to reduce testing costs by over $1M annually through improved quality assessment.

PhD Research Assistant, Purdue University, *West Lafayette, IN, USA* 08/2023 – 08/2027

- Supported commissioning of Herrick Labs by upgrading the building automation system and integrating Modbus.
- Created real-time dashboards for energy/comfort and ML models for occupancy detection (98% accuracy).
- Designed HVAC control strategies with MPC and DeePC, showing DeePC matched MPC without a system model.
- Reviewed 100+ publications on MPC and RL in HVAC, leading to a published review paper in Applied Energy.
- Assisted in developing active current-limiting control for appliances to prevent breaker overloads in electrified homes.

Student Researcher, FortisBC, *Vancouver, BC, Canada* 05/2021 – 08/2023

- Performed assessments on hydrogen injection into natural gas pipelines and renewable gas projects (UBC H_2Lab).
- Worked with engineers and research teams (materials, sensors, AI, safety) to assess industry impacts.

MS Research Assistant, University of British Columbia, *BC, Canada* 05/2021 – 08/2023

- Modeled hydrogen injection into natural gas pipelines with CFD by developing ideal- and real-gas mixture models.
- Produced a journal paper, three conference papers, two awards, and a master's thesis based on the research findings.

Skills

- **Programming:** Python, C, MATLAB, R, MySQL, Git, Linux
- **Machine Learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, Keras
- **Data & Analytics:** NumPy, Pandas, SciPy, Matplotlib, Seaborn, Grafana
- **Modeling & Optimization:** CVX, CVXPY
- **Simulation & Control Testbeds:** OpenAI Gym, OCHRE, BOPTEST
- **Systems & Integration:** Tridium Niagara, InfluxDB, Modbus, IO-Link, MTConnect

Selected Publications — Full list on Google Scholar

- **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze, K.J. Kircher. "Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review." *Applied Energy*, 2025. [DOI]
- **A.J. Khabbazi**, M. Zabihi, R. Li, M. Hill, V. Chou, J. Quinn. "Mixing hydrogen into natural gas distribution pipeline system through Tee junctions." *International Journal of Hydrogen Energy*, 2024. [DOI]
- Six peer-reviewed conference papers (ACM BuildSys, CISBAT, IHPB, CSME, IGEC; 2021–2025).

Selected Honors & Awards

Best PhD Forum Presentation Runner-up Award at ACM BuildSys (2025) — ASHRAE Graduate Student Grant-in-Aid Award (2025) — Best Paper Award at CSME (2023) — Best Presentation Award at CSME (2022) — UBC Graduate Scholarship (2022) — UBC Dean's Entrance Scholarship (2021)

Selected Courses

Artificial Intelligence — Applied Machine Learning — Reinforcement Learning and Control — Intro to Convex Optimization — Industrial IoT Implementation — Statistical Methods — Distributed Energy Resources — Analysis of Thermal Systems — Advanced Thermodynamics